LAB 2

GPIO Programming & ATmega 328P Architecture

Spring 2026

CSE-420L

Embedded Systems Lab

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Section: **C**

"On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work"

Submitted to:

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OBJECTIVE

To interface a push button with **Arduino Uno** and control an LED such that the LED turns ON when the button is pressed and OFF when released.

APPARATUS

- Arduino Uno
- Breadboard
- LED
- Push Button
- 220Ω Resistor
- Jumper Wires
- USB Cable

THEORY

The Arduino Uno is based on the ATmega328P microcontroller.
Digital pins can be configured as:

- INPUT
- OUTPUT
- INPUT_PULLUP

In this circuit:

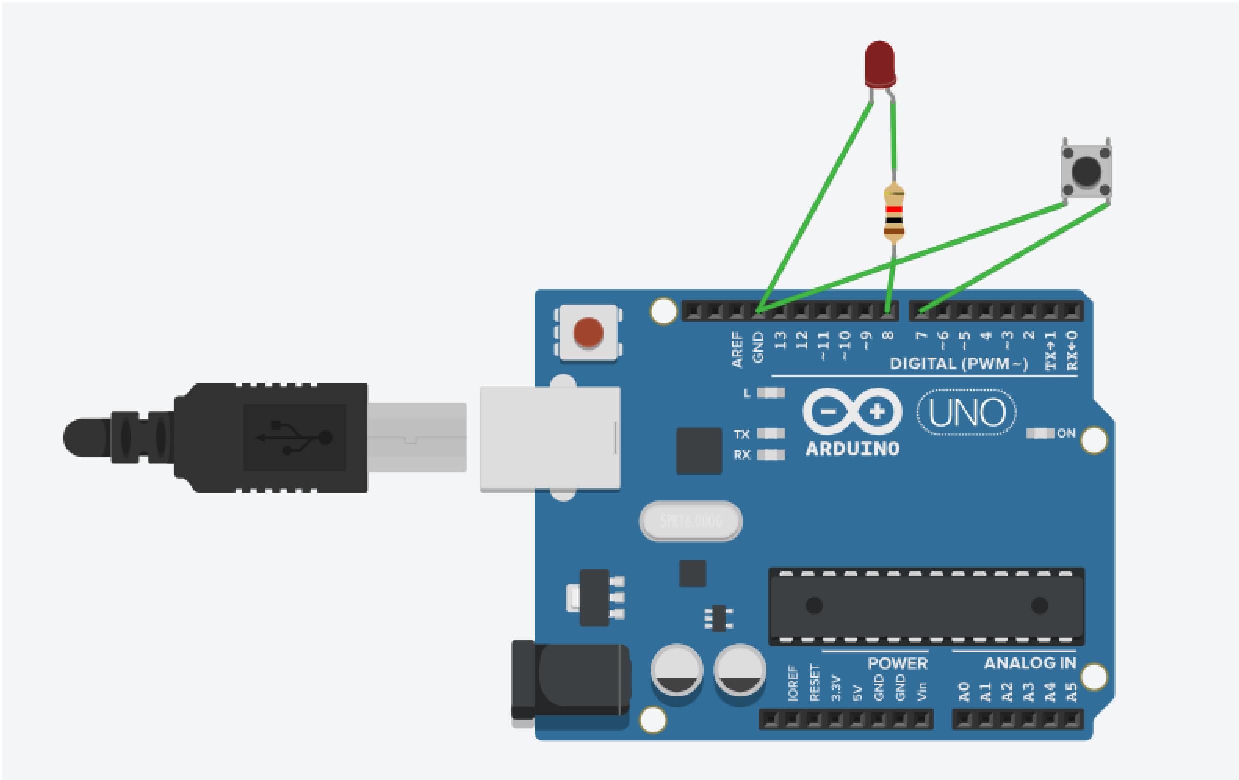
- LED is connected to a digital output pin.
- Push button is connected to a digital input pin.
- Internal pull-up resistor is enabled using INPUT_PULLUP.

When:

- Button NOT pressed → Input reads HIGH
- Button pressed → Input reads LOW

The program checks the button state and controls the LED accordingly.

CIRCUIT DIAGRAM:



CODE:

```
// C++ code

//

void setup()

{

  DDRB |= (1 << DDB0);

  DDRD &= ~(1 << DD7);

  PORTD |= (1 << PORTD7);



}
```

```
void loop()

{

  if(!(PIND & ( 1 << PIND7))){

    PORTB |= (1 << PORTB0);



  }else {

    PORTB &= ~(1 << PORTB0);

  }

}
```

ALGORITHM

1. Start
2. Set LED pin as OUTPUT
3. Set Button pin as INPUT_PULLUP
4. Read button state
5. If button pressed (LOW)
   Turn LED ON
6. Else
   Turn LED OFF
7. Repeat

PROCEDURE

1. Connect Arduino 5V to breadboard positive rail.
2. Connect Arduino GND to breadboard negative rail.
3. Connect LED:
   - Anode → Digital Pin 8
   - Cathode → 220Ω resistor → GND
4. Connect Push Button:
   - One side → Digital Pin 7

    - Other side → GND
5. Upload Arduino code.
6. Press the button and observe LED behavior.

OBSERVATION

Button State	LED State
Not Pressed	OFF
Pressed	ON

The LED glows only when the button is pressed.

CONCLUSION

The push button was successfully interfaced with the Arduino Uno.
The LED was controlled using digital input and output pins.
The experiment demonstrates basic digital I/O operation of Arduino.